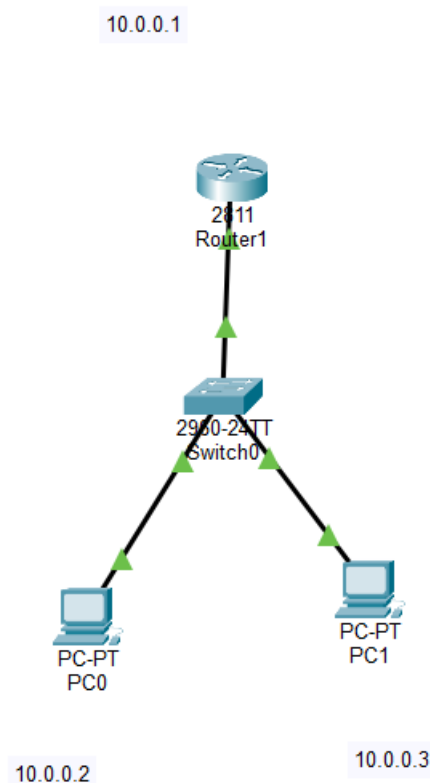


**Day 2: DHCP Configuration Theory (30 min):** DHCP process: Discover, Offer, Request, Acknowledge. **Practical (30 min):** Configure a DHCP server in Cisco Packet Tracer. **Task:** Create a diagram showing the DHCP process flow.

### Task 1.

In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.



| IP Configuration                      |                              |
|---------------------------------------|------------------------------|
| Interface                             | FastEthernet0                |
| IP Configuration                      |                              |
| <input checked="" type="radio"/> DHCP | <input type="radio"/> Static |
| IPv4 Address                          | 10.0.0.3                     |
| Subnet Mask                           | 255.0.0.0                    |
| Default Gateway                       | 0.0.0.0                      |
| DNS Server                            | 0.0.0.0                      |
| IPv6 Configuration                    |                              |

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address: 10.0.0.2

Subnet Mask: 255.0.0.0

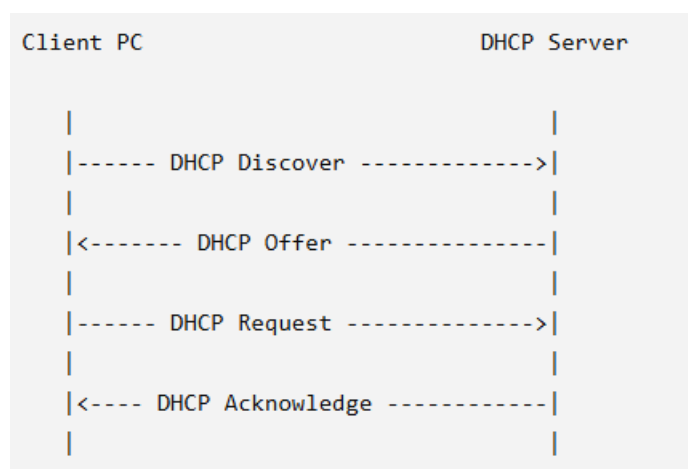
Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

## Task 2.

Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage.  
  
**Hint:** Use arrows to show the direction of each message between client and server.



**Discover:** Client broadcasts asking for an IP address.

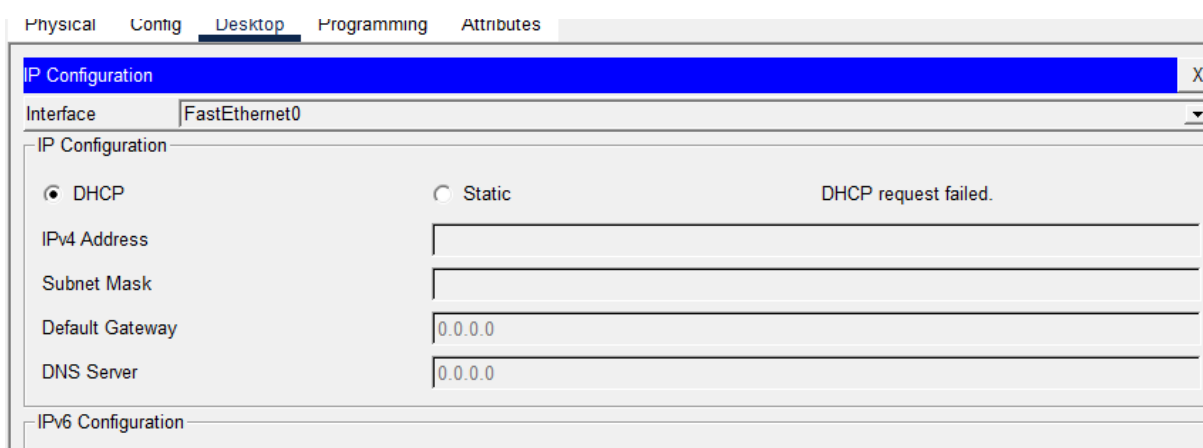
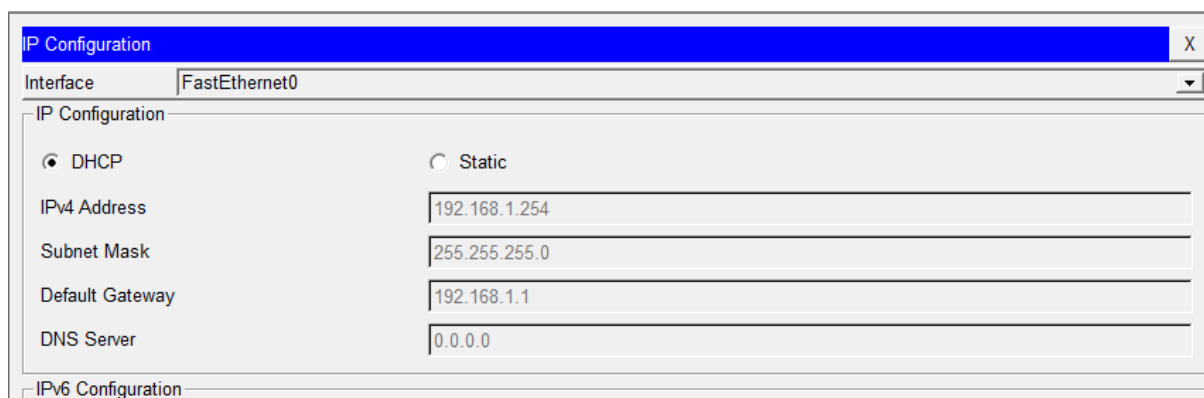
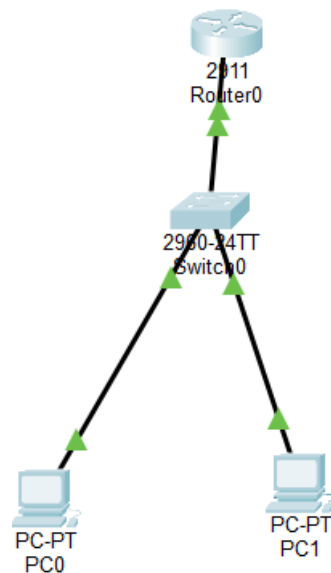
**Offer:** Server offers an available IP address.

**Request:** Client requests the offered address.

**Acknowledge:** Server confirms and leases the IP address.

### Task 3.

Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.



## Result

### Device Result

PC0 Gets IP 192.168.1.254

PC1 DHCP Failed

### Explanation

Only one PC receives an IP address because the DHCP pool contains only one available address. Once it is assigned to the first PC, no additional addresses remain. The second PC cannot obtain an IP and therefore fails to join the network (or receives an APIPA address).

### Task 4.

Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.

### DHCP Configuration

| Setting       | Value         |
|---------------|---------------|
| Pool Name     | Restaurant    |
| Start IP      | 192.168.10.10 |
| Maximum Users | 50            |
| Gateway       | 192.168.10.1  |
| DNS           | 8.8.8.8       |

### Configure Clients

Laptop

Desktop → IP Configuration → DHCP

Generic PC (Smartphone)

Desktop → IP Configuration → DHCP

### **Example Assigned Addresses**

| <b>Device</b> | <b>IP Address</b> |
|---------------|-------------------|
| Laptop        | 192.168.10.10     |
| Smartphone    | 192.168.10.11     |

### **Why DHCP is Useful in a Restaurant**

DHCP automatically assigns IP addresses to customers' devices when they connect to the restaurant's WiFi. This eliminates the need for manual IP configuration, prevents IP address conflicts, and allows many customers to join the network quickly and efficiently.